

Assignment 9

P's utility

$$\begin{aligned} V(\pi - w) &= \pi - w \\ U(w, c_i) &= w - c_i \\ \pi_h &> \pi_l, \quad c_1 < c_2, \quad \gamma_1 < \gamma_2 \\ \bar{u} &= 0 \end{aligned}$$

a)

$$\begin{aligned} \max_{a, w_H, w_L} \quad & \gamma_a \cdot (\pi_H - w_H) + (1 - \gamma_a)(\pi_L - w_L) \\ \text{s.t.} \quad & \gamma_a \cdot (w_H - c_a) + (1 - \gamma_a)(w_L - c_a) = \gamma_a \cdot w_H + (1 - \gamma_a)w_L - c_a \geq 0 \end{aligned}$$

Fix a and set up lagrangian

$$\mathcal{L} = \gamma_a \cdot (\pi_H - w_H) + (1 - \gamma_a)(\pi_L - w_L) + \lambda[\gamma_a \cdot w_H + (1 - \gamma_a)w_L - c_a]$$

Obj and constraints are both linear and thus concave. Which means that FOC are sufficient for optimal solution

PC must be binding otherwise P simply lowers wages overall thus

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial w_H} &= -\gamma_a + \lambda \cdot \gamma_a = 0 \\ \frac{\partial \mathcal{L}}{\partial w_L} &= -(1 - \gamma_a) + \lambda \cdot (1 - \gamma_a) = 0 \\ &\Rightarrow \gamma = 1 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= 0 \Leftrightarrow \gamma_a \cdot w_H + (1 - \gamma_a)w_L = c_a \end{aligned}$$

Thus we simply need

$$\gamma_a \cdot w_H + (1 - \gamma_a)w_L = c_a$$

A solution would be

$$w_H = w_L = c_a$$

Thus P chooses action a_1 if and only if

$$\begin{aligned} \gamma_1 \cdot \pi_H + (1 - \gamma_1) \cdot \pi_L - c_1 &\geq \gamma_2 \cdot \pi_H + (1 - \gamma_2) \cdot \pi_L - c_2 \\ \Leftrightarrow (\gamma_1 - \gamma_2)\pi_H + (\gamma_2 - \gamma_1)\pi_L - c_1 + c_2 &\geq 0 \\ \Leftrightarrow (\gamma_1 - \gamma_2)(\pi_H - \pi_L) &\geq c_1 - c_2 \end{aligned}$$

2)

P sells the firm for a price K such that

$$w_i = \pi_i - K$$

This means that A's optimisation problem becomes P's old one

$$\max \gamma_a \pi_H + (1 - \gamma_a)\pi_L - c_a - K$$

Same as before we see when a_1 is better

$$\gamma_1 \pi_H + (1 - \gamma_1)\pi_L - c_1 - K \geq \gamma_2 \pi_H + (1 - \gamma_2)\pi_L - c_2 - K$$

$$\Leftrightarrow \gamma_1 \cdot \pi_H + (1 - \gamma_1) \cdot \pi_L - c_1 \geq \gamma_2 \cdot \pi_H + (1 - \gamma_2) \cdot \pi_L - c_2$$

$$\Leftrightarrow (\gamma_1 - \gamma_2)(\pi_H - \pi_L) \geq c_1 - c_2$$

Which is the exact condition from earlier

The value of K is the exact value K such that

$$\max_a \gamma_a \pi_H + (1 - \gamma_a) \pi_L - c_a - K = 0$$